

Harness RFQ

Ten low-voltage cable assemblies for a domestic kitchen appliance (a countertop soda dispenser), currently at prototype. Every assembly lands on the main board at one end and on a valve, motor, sensor or display at the other. This document is the complete build package: materials, workmanship, board-side contact order, and a pin-level wire list for each of the ten.

Nothing here is proprietary; ask about anything that is ambiguous rather than assuming.

Scope

In: ten assemblies — J1, J2, J3, J4, J5, J6, J7, J9, J11, J13.

Out: the J10 12 V inlet (2 × 16 AWG, ferrules, landed on the main board) and every AC mains run (AC-1...6, 16 AWG + 18 AWG SJOOW), built in place per `wiring.md`.

Everything in this RFQ is 22 AWG or 28 AWG.

The WAGO fan-outs are built in-house

The five branch nodes in the table below are **excluded from the quote**. A vendor supplies each affected assembly as its trunk only — every board-side conductor run to its full length and terminated at the device end — and the lever nut and its branch legs are made off here. A 221 lever nut takes no crimp and no tooling, and on the one assembly priced both ways the fan-out was 45 % of the assembled cost.

So: quote the wire lists below as written, and ignore the — ... branch rows.

Build quantity

Quote **25, 50, and 100 sets** (a set = all ten assemblies).

Workmanship standard

IPC/WHMA-A-620, Class 2. Every assembly 100% continuity- and isolation-tested pin-to-pin against its wire list below, including the deliberate non-connection at J2 contact 3.

Materials

Item	Spec	Notes
Signal / actuator wire	22 AWG stranded tinned-copper silicone , 600 V, black , 1.7 mm ± 0.1 OD	all ten assemblies except J3 — see Open question 1
Ribbon	28 AWG 4-conductor jacketed ribbon, black	J3 only
Board-end housings	JST XH , 2.5 mm pitch, female crimp housing + contacts	XHP- <i>n</i> per assembly; "XH2.54" is the series name, the pitch is 2.50 mm
Device-end tabs	Female Faston disconnects, 6.3 mm and 4.8 mm	at 20–24 AWG these are TE 2178438-1 (250 / 6.3 mm) and 170214-2 (187 / 4.8 mm); per-device size map not yet recorded
Screw landings	Insulated bootlace ferrules, DIN, 22 AWG	J5 relay terminals, J9 display terminals
Fan-out splices	WAGO 221-420 (10-way) and 221-415 (5-way)	not in scope — we buy and land these ourselves , see below
Sleeve		both cut ends finished with black heat-shrink

Item	Spec	Notes
	Black PET expandable braided, 1/4" / 1/2" / 3/4" per assembly	
Ties	Black UV-nylon, flush-cut, no proud tail	

Every conductor is black. Identification is **by assembly name at the housing** (heat-shrink marker), not per-conductor. Do not substitute a color code.

Where the trunk ends on the five branched assemblies

Five assemblies terminate a shared rail in a WAGO 221 lever nut at the **device end**, not at the main board: one rail conductor rides the trunk and explodes at the manifold or reservoir. **None of that is in this quote** — you supply the rail conductor cut to the length in its wire list and ferruled, and it ends there. We buy the lever nuts, cut the branch legs, and land them here.

The table is here only so the rail conductor's destination is unambiguous. It is not a build instruction.

Assembly	Nut	Ways used	Feed	Branches
J1 MANIFOLD A	221-420	9 of 10	COM	8 valve +
J2 MANIFOLD B	221-415	5 of 5	COM	V-I, V-J, fan, V-K +
J4 SENSORS	221-415	4 of 5	GND	1-wire bus, flow meter, moisture plate
J6 REEDS A	221-415	5 of 5	GND	4 reeds
J7 REEDS B	221-420	7 of 10	GND	6 reeds

Purchased parts, per set

One set = all ten assemblies. Counts are derived from the wire lists below and exclude the fan-outs. Every conductor end appears exactly once: 16 Faston + 13 ferrule + 24 flying = 53, which is the contact count.

#	Part	Spec / MPN	Qty per set	Unit
1	XH housing, 4-way	JST XHP-4 , female	5	ea
2	XH housing, 5-way	JST XHP-5 , female	1	ea
3	XH housing, 6-way	JST XHP-6 , female	1	ea
4	XH housing, 7-way	JST XHP-7 , female	2	ea
5	XH housing, 9-way	JST XHP-9 , female	1	ea
6	XH crimp contact	JST SXH-001T-P0.6	53	ea
7	Faston 250 receptacle	TE 2178438-1 , 6.3 mm, female, 20–24 AWG	12	ea
8	Faston 187 receptacle	TE 170214-2 , 4.8 mm, female, 20–24 AWG	4	ea
9	Bootlace ferrule, insulated	DIN 46228-4, to suit final gauge	13	ea
10	Hook-up wire	black silicone 600 V, 22 AWG, 1.7 mm ± 0.1 OD	17.7	m
11	Ribbon, 4-conductor	28 AWG jacketed, black	1.0	m
12	Braided sleeve, 3/4"	black PET expandable	0.30	m
13	Braided sleeve, 1/2"	black PET expandable	1.70	m
14	Braided sleeve, 1/4"	black PET expandable	2.55	m
15	Heat-shrink, sleeve ends	black, sized to sleeve	18	ends

#	Part	Spec / MPN	Qty per set	Unit
16	Heat-shrink marker, label	black, one per assembly, printed with the assembly name	10	ea

Wire length by assembly (sum of conductor cut lengths, service loop excluded): J1 3050, J2 1750, J4 2200, J5 400, J6 2100, J7 2600, J9 1600, J11 2400, J13 1600 mm. J3 is the ribbon, quoted by the metre instead.

Items 7 and 8 split on an assumption — see Open question 2. Item 9's barrel is sized for 22 AWG; if Open question 1 moves the gauge, it moves with it.

Not purchased by you: the WAGO 221-415 / 221-420 lever nuts and every branch conductor downstream of them. Those are ours.

Board-end contact order

Read contact order from this table, not from conductor names elsewhere. Two connectors are numbered opposite to how their signals read: J1 runs OUT8 → COM , and J6 runs GND , RA4 ... RA1 . Contact 1 is the pin-1 end of the housing.

Conn.	Housing	1	2	3	4	5	6	7	8	9
J1 MANIFOLD A	XHP-9	OUT8	OUT7	OUT6	OUT5	OUT4	OUT3	OUT2	OUT1	COM
J2 MANIFOLD B	XHP-6	COM	FAN	empty	OUT3	OUT2	OUT1			
J3 FAUCET	XHP-4	GND	V5	I035	I033					
J4 SENSORS	XHP-7	3V3	GND	V5	I025	I026	I027	I023		
J5 RELAYS	XHP-4	GND	V5	I02	I019					
J6 REEDS A	XHP-5	GND	RA4	RA3	RA2	RA1				
J7 REEDS B	XHP-7	RB1	RB2	RB3	RB4	CL0	CHI	GND		
J9 DISPLAY	XHP-4	B	A	GND	V12					
J11 GAS	XHP-4	GND	V5	D0UT	A0UT					
J13 PUMPS	XHP-4	AM2	AM1	BM2	BM1					

J2 contact 3 is deliberately empty

J2's wafer is 6-way; only five conductors are populated. OUT4 is a routed spare that drives no valve. **Crimp five contacts into an XHP-6 in their labelled positions and leave contact 3 empty. Do not close the gap.** Contact 3 is mid-housing, not on an end — a loom filling 1–5 consecutively lands every conductor one position off, which puts the shared 12 V rail on a driver output. Verify the empty cavity at final test.

J4 and J7 share a housing

Both are XHP-7. A swap would put J4's 3V3 / V5 onto J7's reed inputs. **Label both at the housing.**

Wire lists

Lengths are conductor cut lengths from the main-board contact to the device termination, service loop excluded — add your standard allowance. Each is measured off the placed CAD assembly, board centre to device centre, scaled by a routed factor and rounded up to the next 50 mm. No enclosure has been built yet, so treat them as tight rather than final and tell us if your process wants more allowance.

J1 — MANIFOLD A · XHP-9 · 3/4" sleeve

Contact	Signal	To	Length	Termination
1–8	OUT8 ... OUT1	valves V-H...V-A, one each	~350 mm	female Faston
9	COM	221-420 at the manifold	~250 mm	ferrule
—	8 × COM branch	221-420 → each valve +	~150 mm	ferrule / female Faston

Each OUT reaches its own coil; COM stops 250 mm out at the lever nut and fans out from there. Low-side switching: COM carries shared 12 V to every valve + ; each valve – returns on its own OUT .

J2 — MANIFOLD B · XHP-6, 5 populated · 1/2" sleeve

Contact	Signal	To	Length	Termination
1	COM	221-415 at the manifold	~400 mm	ferrule
2	FAN	condenser fan –	~450 mm	female Faston
3	—	empty, do not populate	—	—
4	OUT3	valve V-K – , which stands against the board's own flank	~100 mm	female Faston
5	OUT2	valve V-J –	~400 mm	female Faston
6	OUT1	valve V-I –	~400 mm	female Faston
—	4 × COM branch	221-415 → V-I, V-J, fan, V-K	150 / 150 / 100 / 200 mm	ferrule / female Faston

All four branches leave the 221-415 at the manifold, 400 mm along the trunk.

J3 — FAUCET · XHP-4 · 28 AWG 4-conductor ribbon

Contact	Signal	To	Length	Termination
1	GND	faucet display GND	~1 m	per display, see Open questions
2	V5	faucet display 5 V	~1 m	"
3	I035	display TX	~1 m	"
4	I033	display RX	~1 m	"

Straight-through 4-conductor ribbon up the umbilical, no branch. TTL UART; ESD clamping is on the main board, so no cable-end component is required.

J4 — SENSORS · XHP-7 · 1/4" sleeve · three-way branch

Contact	Signal	To	Length	Termination
1	3V3	1-wire temp bus	~300 mm	per device
2	GND	221-415 on the –X wall aft	~300 mm	ferrule
3	V5	flow meter	~300 mm	per device
4	I025	flow meter pulse	~300 mm	per device
5	I026	1-wire data	~300 mm	per device
6	I027	moisture sensor DO	~350 mm	per device

Contact	Signal	To	Length	Termination
7	I023	moisture sensor switched VCC	~350 mm	per device
—	3 × GND branch	221-415 → 1-wire, flow meter, moisture	short pigtails	ferrule

All three legs land within 50 mm of each other; the trunk barely branches by length.

J5 — RELAYS · XHP-4 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1	GND	relay #1 and #2 GND	~100 mm	ferrule, teed to both
2	V5	relay #1 and #2 VCC	~100 mm	ferrule, teed to both
3	I02	relay #2 IN	~100 mm	ferrule
4	I019	relay #1 IN	~100 mm	ferrule

V5 and GND tee at the relay screw terminals — **no WAGO on this assembly.**

J6 — REEDS A · XHP-5 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1	GND	221-415 at reservoir A	~300 mm	ferrule
2–5	RA4 ... RA1	reservoir A level reeds 4...1	~450 mm	reed leads
—	4 × GND branch	221-415 → each reed	short pigtails	ferrule

Note the reversed order: contact 2 is RA4 , contact 5 is RA1 .

J7 — REEDS B · XHP-7 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1–4	RB1 ... RB4	reservoir B level reeds 1...4	~400 mm	reed leads
5	CL0	carbonator low reed	~350 mm	reed leads
6	CHI	carbonator high reed	~350 mm	reed leads
7	GND	221-420 at the cold-core end	~300 mm	ferrule
—	6 × GND branch	221-420 → each reed	short pigtails	ferrule

Same XHP-7 as J4 — label at the housing.

J9 — DISPLAY · XHP-4 · 1/2" sleeve

Contact	Signal	To	Length	Termination
1	B	display RS485 B	~400 mm	ferrule
2	A	display RS485 A	~400 mm	ferrule
3	GND	display 7–36 V input GND	~400 mm	ferrule
4	V12	display 7–36 V input +	~400 mm	ferrule

A and B twisted pair over the run; V12 / GND ride the same sleeve.

J11 — GAS · XHP-4 · 1/4" sleeve

Contact	Signal	To	Length	Termination
1	GND	MQ-6 GND	~600 mm	per device
2	V5	MQ-6 VCC	~600 mm	per device
3	DOUT	MQ-6 digital trip	~600 mm	per device
4	AOUT	MQ-6 analog	~600 mm	per device

Straight-through, no branch.

J13 — PUMPS · XHP-4 · 1/2" sleeve

Contact	Signal	To	Length	Termination
1	AM2	pump A motor tab 2	~400 mm	female Faston
2	AM1	pump A motor tab 1	~400 mm	female Faston
3	BM2	pump B motor tab 2	~400 mm	female Faston
4	BM1	pump B motor tab 1	~400 mm	female Faston

Two differential H-bridge pairs, no shared rail and no fan-out. The pumps ride the pump cartridge, which withdraws from the enclosure, so these four run **unbroken** from housing to motor tab, and the length must hold with that cartridge drawn fully out — do not shorten.

Open questions for the vendor

1. **Wire gauge into the XH housings — 22 AWG silicone, and we want your read.** JST rates the XH contact (SXH-001T-P0.6) #28 to #22 on the conductor and **0.9 to 1.9 mm on the insulation OD**. The silicone we run measures 1.7 mm ± 0.1 at 22 AWG, so it is inside that window with roughly 0.1 mm to spare at the top of the band. Some crimp-selection databases refuse the pair anyway on their own margins, accepting 22 AWG into XH on UL 1007, UL 1015, UL 1061 and ThermoThin and silicone only from 24 AWG down. **Quote 22 AWG silicone. If your process data will not accept it, say so and quote what you would build instead** — 24 AWG silicone, or 22 AWG thin-wall. Current draw on these looms is negligible except MANIFOLD A COM, which carries ≤ ~1.4 A over ~300 mm.
2. **Faston sizes.** `cable-assemblies.md` calls for both 6.3 mm and 4.8 mm disconnects; the per-device map is not yet recorded. Quote against a stated assumption and we will confirm before release.
3. **Device-end terminations on J3, J4, J11.** Sensor and display leads land per device rather than on a standard terminal. Quote these as flying leads, tinned and ferruled, unless you would rather we specify a mating connector.
4. **28 AWG ribbon.** Confirm you can crimp XH contacts onto 28 AWG, or propose the gauge you would rather run in the umbilical.

What we need back

1. **Unit price at 25, 50 and 100 sets**, and any NRE / setup charged per assembly design.
2. **Lead time** for the first article and for a production run.
3. **Your answer to Open question 1** — the wire gauge into the JST XH housings. This is the one open engineering item and we would rather take your recommendation than impose ours.
4. Whether you will build to **IPC/WHMA-A-620 Class 2** and provide continuity + isolation test records per assembly.
5. Anything in the wire lists you would change to make the part cheaper or more manufacturable.

We are quoting several suppliers on the identical package.